Dynamic Trunking Protocol (DTP)

Project Overview

This project was initiated to explore and document the functionality and operational aspects of the **Dynamic Trunking Protocol (DTP)** within Cisco-based switch network environments. The primary objective is to understand how DTP operates, how it facilitates automatic negotiation of trunk links between switches, and how it impacts network configuration and security.

The project focuses on the following goals:

- Understand that by default, all switch ports are configured as access ports and to create a trunk, the port must be manually configured or negotiated via DTP.
- Examine how DTP enables switches to dynamically negotiate their interface roles as trunk or access without manual configuration on both ends.
- Analyze the structure and content of DTP frames, including the interface's DTP mode, accepted encapsulation types, native VLAN, and link status.
- Illustrate DTP operation through an example where a switch configured in dynamic desirable mode negotiates a trunk link with a switch in dynamic auto mode.
- Review the five types of DTP modes: Dynamic Desirable, Dynamic Auto, Trunk, Nonegotiate, and Access, and their behaviors.
- Assess the advantages of DTP such as automatic trunk negotiation, ease of network expansion, and reduction of configuration errors.
- Discuss the disadvantages and security risks including susceptibility to VLAN hopping attacks, unintended trunk formation, incompatibility in mixed vendor environments, and potential network instability.
- Present best practice recommendations such as disabling DTP on access ports, using static trunking, avoiding dynamic modes in sensitive environments, disabling unused ports, and enabling port security on access ports.

The project outcome provides a clear and detailed understanding of how DTP supports network trunk configuration automation while highlighting the importance of secure and deliberate DTP management to maintain network stability and security.

Description

By default, all switch ports are access ports therefore to make a port trunk, the user should manually make it trunk by using DTP. Dynamic Trunking Protocol (DTP) is a Cisco-proprietary protocol designed to simplify the configuration of VLAN trunks between Cisco switches. With DTP, switches can dynamically negotiate their interface roles—either as trunk or access—without requiring manual configuration on both ends.

Recommendations

- Disable DTP on access ports
- Use static trunking
- Avoid dynamic modes
- Disable unused ports
- Use port security on access ports for added protection.

References

https://www.uninets.com/blog/dynamic-trunking-protocol-dtp
https://www.pynetlabs.com/what-is-dtp-dynamic-trunking-protocol/
https://www.nwkings.com/what-is-dtpdynamic-trunking-protocol
https://www.geeksforgeeks.org/computer-networks/dynamic-trunking-protocol-dtp/

Details

DTP operates by the exchange of DTP frames between two neighboring interfaces. The DTP frames include details like:

- The interface's DTP mode
- The interface accepts the encapsulation type
- The interface's native VLAN
- The link's status (up or down)

Let's say an Ethernet wire connects two switches. Switch B's interface is in dynamic auto mode, whereas Switch A's interface is in dynamic desirable mode.
How DTP works in this situation is described by the following steps:

1. Switch A requests the creation of a trunk link by sending switch B a DTP frame.
2. Switch B determines its own mode upon receiving the DTP frame from switch A. It accepts to build a trunk link to switch A since it is configured to operate in dynamic auto mode.
3. Switch B confirms that it wants to create a trunk link by sending a DTP frame to switch A.
4. Receiving the DTP frame from switches B and A confirms that the interfaces agree to create a trunk link.
5. Switch A and switch B establish a trunk link.

There are five types of DTP, or one can say five types of modes in DTP. These are:

- *Dynamic Desirable:* Actively tries to form a trunk. Forms a trunk if the neighbor supports it.
- *Dynamic Auto:* Passive. Becomes a trunk only if the neighbor is in trunk or dynamic desirable. No trunk if both are auto.
- *Trunk:* Forces trunk mode. Can negotiate with neighbor but always sends tagged frames.
- *Nonegotiate:* Disables DTP messages. Use with manually set trunk or access ports.
- *Access:* Forces access mode. Never forms a trunk, DTP is off.

Advantages of Dynamic Trunking Protocol

- DTP automatically negotiates and establishes trunk links between switches, reducing manual setup.
- It allows switches to adjust to changes in network topology without administrator intervention.
- It ensures VLAN traffic is correctly tagged and transmitted, optimizing bandwidth.
- Supports expanding networks by simplifying the addition of new switches or VLANs.
- Minimizes configuration mistakes by automating trunk establishment and VLAN tagging.

Disadvantages of Dynamic Trunking Protocol

- DTP can be exploited in VLAN hopping attacks, as it dynamically creates trunk links that may be used maliciously.
- Neglecting to disable DTP on access ports can lead to unnecessary trunk negotiation and network inefficiency.
- Automated negotiation can lead to unintended trunk links if misconfigured or used in mixed environments.
- DTP is Cisco-specific and may not work seamlessly with non-Cisco devices.
- If DTP is left enabled in environments where it’s not needed, it can lead to network instability.

Affected Locations

- Switches
- Routers
- IP Phones and VoIP Devices
- Wireless Access Points
- Data Centers